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Connectivity in an edge-gateway multipath system

Abstract

In one exemplary aspect, an edge-gateway multipath method includes the step of providing an edge device in a local network communicatively coupled with a cloud-computing service in a cloud-computing network. A set of wide area network (WAN) links connected to the edge device are automatically detected. The WAN links are automatically measured without the need for an external router. The edge device is communicatively coupled with a central configuration point in the cloud-computing network. The method further includes the step of downloading, from the central configuration point, an enterprise-specific configuration data into the edge device. The enterprise-specific configuration data includes the gateway information. The edge device is communicatively coupled with a gateway in the cloud-computing network. The communicatively coupling of the edge device with the gateway includes a multipath (MP) protocol.

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2009/0303880	12/2008	Maltz et al.	N/A	N/A
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2010/0091823	12/2009	Retana et al.	N/A	N/A
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2010/0100768	12/2009	Yamamoto et al.	N/A	N/A
2010/0107162	12/2009	Edwards et al.	N/A	N/A
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2010/0223621	12/2009	Joshi et al.	N/A	N/A
2010/0226246	12/2009	Proulx	N/A	N/A
2010/0290422	12/2009	Haigh et al.	N/A	N/A
2010/0309841	12/2009	Conte	N/A	N/A
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2011/0007752	12/2010	Silva et al.	N/A	N/A
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2011/0078783	12/2010	Duan et al.	N/A	N/A
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2011/0255397	12/2010	Kadakia et al.	N/A	N/A
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2014/0337674	12/2013	Ivancic et al.	N/A	N/A
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Background/Summary

CLAIM OF BENEFIT TO PRIOR APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 16/945,700, filed Jul. 31, 2020, now published as U.S. Patent Publication 2020/0366530. U.S. patent application Ser. No. 16/945,700 is a continuation application of U.S. patent application Ser. No. 15/221,608, filed Jul. 28, 2016, now issued as U.S. Pat. No. 10,749,711. U.S. patent application Ser. No. 15/221,608 is a continuation-in-part of U.S. patent application Ser. No. 14/321,818, filed on Jul. 2, 2014, now issued as U.S. Pat. No. 9,722,815. U.S. patent application Ser. No. 14/321,818 claims priority to U.S. Provisional Patent Application 61/844,822, filed on Jul. 10, 2013. U.S. Patent Publication 2020/0366530, U.S. Pat. Nos. 10,749,711 and 9,722,815, and U.S. Provisional Patent Application 61/844,822 are hereby incorporated by reference in their entirety.

BACKGROUND

(1) Several trends, are altering the use of enterprise applications. For example, enterprises are moving to hosting applications in private and public clouds as opposed to enterprise data centers. Enterprises are also increasingly using applications provided by other companies, which are generically grouped under SaaS (Software-as-a-Service) and are not hosted in an enterprise-data center. In another example, enterprises are migrating from large Information Technology (IT) supported branches to smaller branches. These smaller branches can utilize remote IT management strategies.

(2) These trends have combined to alter application's network paths and/or the quality of service (QoS) of these paths. With enterprise data-center applications, the large IT branches can lease multiprotocol label switching (MPLS) lines. MPLS can be mechanism in communications networks that directs data from one network node to the next node based on short path labels rather than long network addresses, thus avoiding complex lookups in a routing table. MPLS lines can be associated with a known level of QoS that provides a deterministic application access experience

and/or application availability. Applications are moving to the cloud where they are deployed either in the public and/or hybrid cloud. Enterprise branches access these applications via the public Internet. Access to these applications in such cases may be hampered by the 'best effort' nature of access as opposed to having a known QoS level. Additionally, a smaller branch may also utilize computing devices that are relatively easy to deploy and/or remotely manage in the event no on-site IT staff is available.

BRIEF SUMMARY OF THE INVENTION

(3) In one aspect, a network-link method useful for a last-mile connectivity in an edge-gateway multipath includes the step of identifying a network-traffic flow of a computer network using deep-packet inspection to determine an identity of an application type associated with the network-traffic flow. The network-link method includes the step of aggregating a bandwidth from a specified set of network links. The network-link method includes the step of intelligently load-balancing a traffic on the set of network links by sending successive packets belonging to a same traffic flow on a set of specified multiple-network links. The set of specified multiple-network links is selected based on the identity of an application type associated with the network-traffic flow. The network-link method includes the step of identifying a set of active-network links in the set of specified multiple-network links. The network-link method includes the step of providing an in-order data delivery with an application persistence by sending data packets belonging to a same data-packet flow on the set of active links. The network-link method includes the step of correcting an error on a lossy network link using an error-control mechanism for data transmission selectively based on the identified network-traffic flow and a current measured condition in the computer network.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present application can be best understood by reference to the following description taken in conjunction with the accompanying figures, in which like parts may be referred to by like numerals.
- (2) FIG. 1 illustrates an example a programmable, multi-tenant overlay network, according to some embodiments.
- (3) FIG. 2 depicts a process of a network link used to replace 'last mile' connectivity, according to some embodiments.
- (4) FIG. 3 depicts a process of removing the requirement for an IT administrator to configure each individual device in an enterprise computing network, according to some embodiments.
- (5) FIG. 4 illustrates an example flow sequence diagram for an MP packet flow, according to some embodiments.
- (6) FIG. 5 illustrates an example MP process for bandwidth aggregation and data ordering, according to some embodiments.
- (7) FIG. 6 illustrates an example of metadata in an MP header, according to some embodiments.
- (8) FIG. 7 depicts an exemplary computing system that can be configured to perform any one of the processes provided herein.
- (9) The Figures described above are a representative set, and are not an exhaustive with respect to embodying the invention.

DETAILED DESCRIPTION

(10) Disclosed are a network-link method and system useful for a last-mile connectivity in an edge-gateway multipath. Although the present embodiments have been described with reference to specific example embodiments, it can be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the particular example embodiment.

(11) Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

(12) Furthermore, the described features, structures, or characteristics of the invention may be combined in any suitable manner in one or more embodiments. In the following description, numerous specific details are provided, such as examples of programming, software modules, attendee selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments of the invention. One skilled in the relevant art can recognize, however, that the invention may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the invention.

(13) The schematic flow chart diagrams included herein are generally set forth as logical flow chart diagrams. As such, the depicted order and labelled steps are indicative of one embodiment of the presented method. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more steps, or portions thereof, of the illustrated method. Additionally, the format and symbols employed are provided to explain the logical steps of the method and are understood not to limit the scope of the method. Although various arrow types and line types may be employed in the flow chart diagrams, they are understood not to limit the scope of the corresponding method. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the method. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted method. Additionally, the order in which a particular method occurs may or may not strictly adhere to the order of the corresponding steps shown.

(14) Example Definitions

(15) Automatic Repeat reQuest (ARQ) can be an error-control method for data transmission that uses acknowledgements (e.g. messages sent by the receiver indicating that it has correctly received a data frame or packet) and timeouts (e.g. specified periods of time allowed to elapse before an acknowledgment is to be received) to achieve reliable data transmission over an unreliable service. If the sender does not receive an acknowledgment before the timeout, it retransmits the frame/packet until the sender receives an acknowledgment or exceeds a predefined number of re-transmissions

(16) Deep-packet inspection (DPI) can include a form of computer network packet filtering that examines the data part (and also the header in some embodiments) of a packet as it passes an inspection point.

(17) Forward error correction (FEC) can be used for controlling errors in data transmission over unreliable or noisy communication channels. A sender can encode a message in a redundant way by using an error-correcting code (ECC). FEC codes can include block codes, convolutional codes, etc.

(18) Last-mile connectivity can refer to the final leg, of the telecommunications networks delivery components and mechanisms.

(19) Lossy can refer to data compression in which unnecessary information is discarded.

(20) Multiprotocol Label Switching (MPLS) can be a type of data-carrying technique for high-performance telecommunications networks that directs data from one network node to the next based on short path labels rather than long network addresses, avoiding complex lookups in a routing table. The labels can identify virtual links between distant nodes rather than endpoints. MPLS can encapsulate packets of various network protocols. MPLS can support a range of access technologies, including, inter alia: T-carrier (e.g. T1)/E-carrier (E1), Asynchronous Transfer Mode

(ATM), Frame Relay, and Digital subscriber line (DSL).

(21) Quality of service (QoS) can refer to the overall performance of a telephony or computer network, particularly the performance seen by the users of the network.

(22) Software as a Service (SaaS) can be a software licensing and delivery model in which software is licensed on a subscription basis and is centrally hosted.

(23) Virtual Machine (VM) can be an emulation particular computer system.

(24) Link Characterization can refer to measuring the quality of a link which will include the latency (e.g. one-way packet delay), jitter (e.g. packet delay variation), loss (e.g. what percentage of packets are actually delivered at the receiving end) and available bandwidth.

(25) Measured Condition in a computer network can refer to the characterization of one or more links that are connected to an edge device.

(26) Error Control Mechanism can refer to the remedial action taken by an edge device or a gateway device to overcome the side effects of a non-perfect link. These mechanisms can be used to overcome jitter and loss experienced in any one link and include forward error correction (FEC), duplication of packets (e.g. if multiple links are available). This can also include an implementation of a jitter buffer which can minimize the effects of the packet delay variation.

(27) Example Methods And Systems

(28) FIG. 1 illustrates an example of a programmable, multi-tenant overlay network **100**, according to some embodiments. An overlay network can be a computer network built on the top of another network. Overlay network **100** can include a distributed system such as a cloud-computing network (e.g. public cloud **102**). Public cloud **102** can include a cloud-computing network. In some embodiments, public cloud **102** can be implemented, in whole or in part, as private cloud-computing network (e.g. a proprietary network or datacenter that uses cloud computing technologies). In other embodiments, the public cloud **102** can include SaaS companies **109** which provide applications to enterprises and end-consumers. As used herein, a cloud-computing network can include a computer network(s) that utilizes a variety of different computing concepts that involve a large number of computers connected through a real-time communication network (e.g. the Internet). A public cloud can include a set of computers and computer network resources based on the standard cloud-computing model, in which a service provider makes resources, such as applications and storage, available to the general public over the Internet. Applications, storage, and other resources can be made available by a service provider.

(29) Public cloud **102** can include orchestrator **104** (e.g. a Velocloud® orchestrator), Orchestrator **104** can enable configuration and monitoring of the network from any location with Internet access. Orchestrator **104** can be a central controller for configuring and monitoring a multi-tenant instance of the overlay network described by a unique ‘network ID’, Each such instance can have a set of tenant(s) that have tenant specific policies for sharing resources, access control and configuration. A tenant can then have a ‘tenant-id’ which is used to identify tenants in the network. Multiple independent instances of networks can exist so as to enable self-operated overlay networks similar to the public network.

(30) In this context, an orchestrator **104** can perform various functions such as configuration and monitoring. Orchestrator **104** can enable role based configuration and management. The following can be examples of roles. An ‘end-user’ (e.g. maps to an access device like laptop or mobile device) that connects to an edge device **108** that can be enabled to configure and/or monitor resources and policies that are specific to a user. A ‘tenant administrator’ can be a tenant administrator. A tenant administrator can configure tenant-wide policy and by extension policies for all the users in the tenancy. An ‘operator’ that can operate the overlay network by provisioning gateway(s) **106**, edge device(s) **108** and/or other resources for the network (e.g. may not be able to view or modify tenant policies) can be provided.

(31) In addition to this, the orchestrator **104** can also enable ‘authenticated partners’ to modify the behavior of the network (e.g. application service providers who want to reserve extra bandwidth for

some applications sessions etc.) via published application program interface (APIs).

(32) Public cloud **102** can include gateway(s) **106**. A gateway can be a network node equipped for interfacing with another network utilizing different communication protocols. Gateway(s) **106** can be deployed in a public cloud (e.g. as shown in FIG. **1**), a private cloud, Internet service provider (ISP) peering points and/or application service peering points that serve as aggregation points for multiple edges. Gateway(s) **106** can be located at peering points in public cloud **102**.

(33) Edge device **108** can provide entry points into enterprise and/or service-provider core networks. Example edge devices can include routers, routing switches, integrated access devices (IADs), multiplexers, and a variety of metropolitan area network (MAN) and wide area network (WAN) access devices. Edge device **108** can be deployed inline in one of several modes. In one example, edge device **108** can be deployed as a customer premises equipment (CPE) device in a branch that is capable of serving as a router. In one example, edge device **108** can be deployed as a powered mobile device that can be attached to end-user devices (e.g. laptops, desktops, wearable computers, tablet computers and the like via universal serial bus (USB)). In some examples, edge device **108** can include device software that directly interacts with a host-device operating system. In one example, the edge device **108** may be a virtual machine. A virtual machine can be a software-based emulation of a computer. In some examples, edge device **108** and the gateway(s) **106** can straddle the ‘bottleneck’ section of a communication network (e.g. the ‘last-mile’—a final leg of a communication networks delivering communications connectivity to a network host such as an enterprise computing system). In some embodiments, edge device **108** can be characterized as ‘zero touch’ (e.g. no configuration explicitly required at client side). Accordingly, edge device can automatically detect an available wide area network (WAN) and locate orchestrator **104**.

(34) The edge device **108** sends network packets. Network packets may be control packets, data packets or management packets. Control packets or control traffic is used to sense the quality of the path, link characteristics, clock synchronization etc. This is also known as the control plane. Data packets or data traffic are packets can be sent from the client and/or source computer to the application server running in the enterprise data center or private or public cloud **102**. This is also known as the data plane. Management packets or management traffic are packets can be sent from the edge **108** or gateway **106** to the orchestrator **104** and includes heartbeat messages or flow statistics etc. This is also known as the management plane. In one example, both the control plane and the data plane can pass through the gateway **106**. In some examples, only the control traffic may be sent to the gateway **106** and the data plane may bypass the gateway **106** and go directly from the edge **108** to the application server.

(35) FIG. **2** depicts a process **200** of a network link used to replace a ‘last mile’ connectivity (e.g. last MPLS, T1 etc.), according to some embodiments. The network links can be multiple consumer grade broadband links, private links (MPLS, etc.), WiFi networks or 3g/4g mobile links with the ability to perform process **200**. In step **202** of process **200**, network traffic can be identified using deep-packet inspection to determine the application and/or application type of the traffic.

Appropriate measures can be applied to ensure the QoS of the specific traffic based on the application, application type (e.g. real-time, transactional, bulk) and/or business priority of the traffic. For example, if the network traffic is identified as voice traffic which is a high business priority, then forward-error correction can be performed to reduce or eliminate packet loss. In another example, the network traffic can be identified as a bulk-file transfer. In this example, the file-transfer network traffic can be set as the lowest-priority traffic and can use a small portion of bandwidth under contention or more bandwidth if no other traffic is in the network. Traffic identified as ‘regular web browsing’ (such as Facebook® and YouTube®) can be dropped out of the network altogether and sent over the regular Internet as it is not business critical. In step **204**, bandwidth can be aggregated from all the links (e.g. a link can be a communications channel that connects two or more communicating devices). For example, bandwidth can be aggregated with a multipath transport layer protocol capable of ‘striping’ a traffic flow (e.g. flow of data packets from

a source to a destination) across multiple paths between two peers (e.g. edge **108** and/or gateway **106**). Traffic flow can be 'striped' across the multiple paths in one peer and 'gathered' at the other peer. In step **206**, traffic on the links can be intelligently load balanced by sending successive packets belonging to the same flow (e.g. a traffic flow) on multiple links selected by an application aware intelligent link characterization and/or link selection it is noted that the selected QoS based on the application can inform the selected links (e.g. whether to bind traffic to the best link, load balance or replicate traffic, etc.). The selected QoS can also determine whether the application is sensitive to loss and/or jitter. Based on the levels of loss and jitter in the network and the sensitivity of the traffic to them, a mitigation mechanism is put into play. In step **208**, outages can be prevented using reliable, self-correcting data transfer to ensure in-order data delivery with the ability to maintain application persistence, as long as there is at least one active link, by sending packets belonging to the same flow on the active link(s). In step **210** errors on lossy links can be corrected using an error control mechanism for data transmission (e.g. Automatic Repeat-reQuest (ARQ) and/or forward error correction (FEC)) selectively based on the traffic identified and the current measured, conditions in the network.

(36) FIG. **3** depicts a process **300** of removing the requirement for an IT administrator to configure each individual device in an enterprise computing network, according to some embodiments. In step **302**, WAN links that are connected directly to the edge device can be detected and measured without the need for an external router. In step **304**, a central configuration point in the cloud can be connected to. Enterprise-specific configuration data, including available gateways, can be downloaded. In step **306**, the available gateway(s) can be connected to by the entity in the cloud with the enterprise-specific configuration data. In step **308**, an available bandwidth on each path can be measured.

(37) A multipath (MP) protocol can implemented by combining multiple network paths into a composite connection that multiplexes packets from MP packet flows and control information (path quality, link characteristics, clock synchronization, etc.). An MP packet flow can map to a pair of internet protocol (IP) flows (e.g. one flow in each direction such as forward and reverse and between two endpoints). The MP packet flow can be identified by a set of parameters that describe a pair of IP flows (e.g. five (5)-tuple (the reverse path is described with source and destination swapped), namely: source IP address, destination IP address, source port, destination port, and the network layer three (3) protocol identifier. In some examples, a multipath routing can include a routing technique of using multiple alternative paths through a network.

(38) In effect every network layer four (4) flow (e.g. a pair of layer three 3) flows) (e.g. an application flow) can have a one-to-one mapping with the MP packet flow. In some embodiments, an, application flow and MP packet flow refer to the same notion. Each MP packet flow can be assigned an MP packet-flow identifier. The MP packet-flow identifier can be unique to the set of MP peers (e.g. peer one (1) and/or peer two (2) of FIG. **4**). An MP node can aggregate connections from multiple MP peers that are sending MP packet flows to it (e.g. gateway(s) **106**). The MP node can aggregate flow identifiers generated by the non-aggregation peer (e.g. edge device **108**) that may not be unique.

(39) In one example, at the time of first connection between two MP stacks, an MP_INITIATE message can be passed which assigns a unique identifier that is used by the non-aggregation peer to ensure the flow identifier is unique at the aggregation point (see FIG. **4**). In one example, this operation can be implemented to an identifier specific to the particular network of the non-aggregation peer.

(40) FIG. **4** illustrates an example flow sequence **400** diagram for an MP packet flow, according to some embodiments. In one embodiment, peer one (1) can be a client-side edge device and peer two (2) can be a cloud-based gateway device. Peer one (1) can transmit an MP_INITIATE **402** to peer two (2). MP_CONTROL **408** (e.g. control information such as QoS parameters, treatment of data traffic flow parameters, etc.) can be exchanged between peer one (1) and peer two (2). Data packets

can then be exchanged (e.g. MP_DATA **404** and MP_DATA **412**). Data packets can include any user data. These data packets can be sequenced, numbered and/or sent across multiple links. When sent across multiple links, redundant copies of the packets can be purged on receipt. Data packets can be acknowledged on return. Additional control data (e.g. MP_CONTROL) can be exchanged. MP_FIN **406** can initiate closing the MP packet flow session by peer one (1). Peer two (2) can provide MP_FIN+ACK **412** to acknowledge MP_FIN **406** can terminate session.

(41) FIG. 5 illustrates an example MP process **500** for bandwidth aggregation and data ordering, according to some embodiments. Process **500** can be used for intersession load balancing. In some embodiments, the MP stack can achieve bandwidth aggregation by sending successive packets belonging to the same MP packet flow on the different paths, to a peer MP stack. In some examples, different paths can be established on different links (though this is not a limiting condition). For example, data packets **502** can be an application flow. Data packets **502** can be striped with a MP stripe **504** in one device (e.g. edge device **108**).

(42) The endpoints (e.g. client and the application server) can infer this as an aggregated throughput as more packets are delivered to the endpoints in a given time window when compared to non-multipath case. MP process **500** can deliver ordered data **505** and **508** between two MP peers even if the data is sent on different paths between the peers. For example, successive data packets belonging to the same flow can be sent on different links with additional metadata. The metadata can identify data packet absolute offsets in the original flow. This metadata can be used to re-arrange the data back in order when the underlying application requires in-order data. In some applications (e.g. real-time collaboration applications) this re-ordering may introduce latencies that may be unacceptable. In these instances, data packets can be delivered in the same order of arrival. The application can handle ordering of data packets. This application awareness can be in the transport layer. This presence can be implemented on both sides of the network and enable interpretation of metadata and reassemble the data. This functionality can be selectively turned on/off based on detecting an application's particular requirements on receiving the ordered data **506** and **508**. Additional headers, shown below, marked with MP headers **510** and **514** (e.g. "Vn") can be added. MP headers **510** and **514** can describe the data ordering along with other metadata (e.g. such as MP packet flow identifier, timestamps).

(43) FIG. 6 illustrates an example of metadata in an MP header, according to some embodiments. This metadata can enable the peer MP stack to receive the MP packet flows **506** and **508** (including striped data packets **512** and **515**) from different paths in their order of arrival and re-arrange them in order to re-create the original flow of data packets **502** as data packets **520**.

(44) In one example, a Global Data Sequence Number (GDSN) can be the byte offset of the data with respect to the original application flow (e.g. data packets **502**). GDSN can be used to reorder the data. Each MP packet can have the GDSN which is used by the peer MP stack to re-order the MP packet flow in an original order. Additionally, each peer can transmit the last seen GDSN on its stack for a given MP packet flow 'piggybacked' on an MP data packet. This last seen GDSN can be used to purge queues and re-transmit a missing GDSN. In the case the data transfer is half-duplex, then a MP_ACK message can be explicitly used to transmit the last seen GDSN to the other peer.

(45) In the context of FIG. 4, during the closing of the MP packet flow **400**, the MP_FIN **406** can be set by the peer that initiates the closing of the MP packet flow **400**. The GDSN in this packet can be used by the other peer to acknowledge teardown via the MP_FIN+ACK **414** with a GDSN of zero (0).

(46) An example method of traffic identification is now provided. An MP system can utilize an external deep-packet inspection engine (and/or form of computer network packet filtering) to identify the application and application type of a given flow. This information can be used to determine the optimal MP packet flow settings to ensure the MP packet flow's QoS parameter. In cases where the application cannot be identified, an MP system can monitor the behavior of MP packet flows over time and attempt to derive the optimal settings for QoS. Future flows can be

mapped to these new settings using IP address, port number, protocol, TOS/DSCP tag and/or destination hostname as the MP system learns optimal MP traffic parameters. Additionally, these settings which were obtained through this slow learning method (e.g. can include machine-learning methodologies such as neural networks, supervised learning, unsupervised learning, clustering, structured prediction, decision tree learning, reinforcement learning and the like) can be shared to all other edges in the network via the orchestrator **104** which can allow learning to, be spread across an enterprise or the entire network of connected edges.

(47) Deep-packet inspection (DPI) can include examining, the data part (and/or also the packet header, etc.) of a packet as it passes an inspection point, searching for protocol non-compliance, viruses, spam, intrusions, or defined criteria to decide whether the packet may pass or if it needs to be routed to a different destination, or, for the purpose of collecting statistical information. DPI can be performed with a DPI engine (e.g. Qosmos®, DPI engine, Dell™ SonicWALL™ Reassembly-Free Deep-Packet Inspection™ (RFDPI) engine, etc.) and/or other packet analyser.

(48) An example of path characterization and selection is now provided. An MP protocol can implicitly implement communicating an MP packet flow on multiple paths (e.g. on a set of underlying links). Consequently, an active path characterization that periodically measures the health of a path can be implemented. A cost function that computes a cost for each path based on the parameters measured in the health check can be implemented. A selection algorithm can be implemented that applies a set of constraints and chooses the path based on the path cost and the transmit algorithm determined.

(49) An example of active path characterization is now provided. As a part of link characterization, the latency (e.g. one-way packet delay), jitter (e.g. packet delay variation), loss and available bandwidth on the path can be measured. To measure latency between two MP peers on a given path, a clock synchronization operation can be implemented in the MP peers. An example time synchronization protocol is now provided. Timestamp measurements can be sent continuously to whichever device is performing the role of master clock. The lowest difference in timestamps from a set of measurements can be used as a measure of the offset between the clocks. Backward time shifts which could influence measurements and computation can be avoided. The drift rate can be measured by observing the change in offset over time. Based on this drift rate, the interval between continuous measurements can be calculated to ensure that clocks will remain synchronized over time. Once the docks are synchronized, the one-way receive latency and jitter can then be measured by sending a timestamped packet train.

(50) In one example, multipath transport can handle and/or prevent congestion issues when the network paths are sufficiently diverse from a network topology standpoint. In these cases, the overall load on the individual paths can be reduced. On the other hand, diverse network paths can have diverse, characteristics in terms of latency, throughput, loss and/or jitter. The load-balancing algorithm can send packets are, on a 'best possible' link until the point the link is oversubscribed and/or there is loss on the link before switching to another path. When the network includes a wireline backbone (cable, DSL, etc.), alternate paths can be utilized when available. On the other hand, with respect to networks with a wireless backbone (e.g. mobile, WiFi, WiMax, etc.), a packet drop may be an 'ephemeral' event that is short lived with relatively quicker recovery. In such a case, it may not be prudent to switch to alternate paths or clamp down the rate for this event without consideration of various other metrics. Thus, other metrics in addition to a loss value can be utilized. For example, a combination of parameters can be utilized, including, inter alia: the ECN flag (e.g. explicit congestion notification) set by an upstream router in an IP layer, a rate of acknowledgements received, a rate of loss in an interval of time to estimate the lossy value of a link, etc.

(51) In one example, the cost of a MP path can be computed as the ϵ taken for a data packet to reach from one peer to another peer inclusive of such factors as scheduling and/or MP processing overheads. It can be computed as the sum of the jitter, latency and processing delays. The path with

the least cost with respect to, a given set of constraints (e.g. link level policies, application specific path usage policies etc.) can be, selected accordingly.

(52) An example MP path selection method is now provided. Based on the application and/or the current measured network conditions, the MP path(s) can be treated in various ways. In one example, the MP path can be load balanced such that each data packet selects the path with the lowest path cost and is transmitted on that path. In another example, the MP path can be fixed such that the first packet selects the best path available. This path can be used as long it is available. If an active path characterization operation determines that the path is no longer available, a path selection operation can be performed again and the MP packet flow can migrate to a next best path. In yet another example, the MP path can be replicated across n-number paths based on such parameters as, inter alia: the importance of the application, bandwidth required and/or expected potential for packet loss in the network.

(53) In one example, QoS can be ensured for an application by utilizing a combination of path selection methods such as those provided supra, as well as, network scheduling, packet reordering and/or error correction. For example, when an MP packet flow is initiated, an edge device (e.g. edge device **108**) can identify the application and determine the proper QoS methods to be applied for this type of flow. The methods may or may not be symmetric (e.g. the same for the sender and receiver). Once the edge device determines the methods to be used, a control message can be sent to the gateway to ensure that the gateway (e.g. gateway(s) **106**) in turn has information as to how to treat the MP packet flow (e.g. without having to do its own application identification). In the event the MP system (e.g. based on network conditions) and/or an administrator indicates that the methods should be changed, the edge device can again signal the gateway with a control message. The methods can be updated without interruption to the service or traffic. For example, upon receipt of the control message from the edge, the gateway can update the QoS methods of the flow without deleting the existing flow. As a result, the next packet to be sent can, use the updated scheduling policies and link, selection methods that were transmitted without interruption. For example, a MP packet flow that is being load, balanced and is changed to replication as loss increases in the network can load balance packets 1–n until the control message is received. Packet flow can be a sequence of packets from a source computer to a destination, which may be another host, a multicast group, or a broadcast domain. Accordingly, packets beginning with n+1 can begin to be replicated.

(54) The gateway can be a multi-tenant gateway wherein multiple customers with edge devices can connect to the same gateway without actually exposing any of their data to each other. The multi-tenant gateway can implement a two-level hierarchical scheduler. In this case, a total egress bandwidth to the edge can be equally shared (e.g. in a work conserving manner) between all the connected edges at the top level (e.g. root level). The second level (e.g. a leaf) can schedule the MP packet flows belonging to a particular edge device rather than have resource limits defined for that edge device by the top level. To ensure that the scheduler does not hit processing limits for scheduling flows, the leaf level scheduler may not have per flow queues. Instead, a multiplexing algorithm can be utilized. The multiplexing algorithm can map a set of flows characterized by a set of parameters to a set of queues such that there is a many to one mapping between flows and queues.

(55) In one example, SaaS applications may also directly interact with the software in the edge device (e.g. edge device **108** in FIG. 1) or gateway(s) (e.g. gateway **106** in FIG. 1). This can be done, for example, to query the health of the last-mile and to provision network bandwidth and characteristics in the last-mile to ensure QoS for the application. The edge device and/or the gateway(s) regardless of their embodiments provide APIs (application programming interfaces) that a SaaS application, with suitable permissions, can use to determine how the last-mile from the edge device to the gateway (e.g. in both directions) is performing. With this information the SaaS application may throttle back bandwidth so that the application continues to operate without

congesting the network further and yet function reasonably well. By default, when an end-user accesses an application (which may run anywhere in the public cloud), the edge device identifies the application and determines the proper QoS methods to apply for this type of flow. As noted in supra, this includes network scheduling, packet reordering and/or error correction which is determined by policies set in the Orchestrator (e.g. orchestrator **104** in FIG. **1**). The SaaS application may modify these policies dynamically to ensure that the end-user gets the best experience possible given the current last-mile characteristics.

(56) FIG. **7** depicts an exemplary computing system **700** that can be configured to perform any one of the processes provided herein. In this context, computing system **700** may include, for example, a processor, memory, storage, and I/O devices (e.g., monitor, keyboard, disk drive, Internet connection, etc.). However, computing system **700** may include circuitry or other specialized hardware for carrying out some or all aspects of the processes. In some operational settings, computing system **700** may be configured as a system that includes one or more units, each of which is configured to carry out some aspects of the processes either in software, hardware, or some combination thereof.

(57) FIG. **7** depicts computing system **700** with a number of components that may be used to perform any of the processes described herein. The main system **702** includes a motherboard **704** having an I/O section **705**, one or more central processing units (CPU) **708**, and a memory section **710**, which may have a flash memory card **712** related to it. The I/O section **706** can be connected to a display **714**, a keyboard and/or other user input (not shown), a disk storage unit **716**, and a media drive unit **718**. The media drive unit **718** can read/write a computer-readable medium **720**, which can include programs **722** and/or data.

(58) B. Conclusion

(59) Although the present embodiments have been described with reference to specific example embodiments, various modifications and changes can be made to these embodiments without departing from the broader spirit and scope of the various embodiments. For example, the various devices, modules, etc, described herein can be enabled and operated using hardware circuitry, firmware, software or any combination of hardware, firmware, and software (e.g., embodied in a machine-readable medium).

(60) In addition, it can be appreciated that the various operations, processes, and methods disclosed herein can be embodied in a machine-readable medium and/or a machine accessible medium compatible with a data processing system (e.g., a computer system), and can be performed in any order (e.g., including using means for achieving the various operations). Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. In some embodiments, the machine-readable medium can be a non-transitory form of machine-readable medium.

Claims

1. A method for directing a packet flow along a plurality of network links connecting a site to a wide area network (WAN), the method comprising: at the site: receiving different packets of a packet flow; performing deep packet inspection (DPI) on the received packet flow to identify an application associated with the received packet flow; and based on the identified application, selecting different network links from the plurality of network links for forwarding different sets of packets in the packet flow through the WAN, wherein (i) said different network links comprise at least two of a broadband link, a private multiprotocol label switching (MPLS) link, and a wireless link, (ii) each network link connecting the site to the WAN, and (iii) the selection of the different network links for the different sets of packets in the packet flow allowing the packets of the packet flow to take different paths to reach a destination outside of the site through the WAN; wherein the selection of the different network links comprises reordering out-of-sequence packets upon arrival

at a destination device using a global data sequence number (GDSN), and error correction being performed by retransmitting missing packets based on a last-seen GDSN value exchanged between peers.

2. The method of claim 1, wherein an edge device at the site selects the different network links.

3. The method of claim 1 further comprising aggregating bandwidth from the plurality of network links with a multipath transport layer protocol.

4. The method of claim 3, wherein the multipath transport layer protocol performs the selection of the different network links for the different sets of packets in the packet flow.

5. The method of claim 4, wherein the multipath transport layer protocol uses an identity of the application to perform the selection.

6. The method of claim 1, wherein said selecting comprises load balancing the different sets of packets across different selected links.

7. The method of claim 6, wherein the load balancing is an application-aware load balancing.

8. The method of claim 6, wherein load balancing traffic comprises assessing a quality of a network link based on an application type of the identified application by measuring a quality of each network link based on at least one of latency, jitter, loss, and available bandwidth.

9. The method of claim 1, wherein the different selected network links comprises all network links in the plurality of links.

10. The method of claim 1, wherein the different selected network links do not include all the network links in the plurality of links.

11. A non-transitory machine readable medium storing a program for execution by a set of processing units, the program for directing a packet flow along a plurality of network links connecting a site to a wide area network (WAN), the program comprising sets of instructions for: receiving different packets of a packet flow; performing deep packet inspection (DPI) on the received packet flow to identify an application associated with the received packet flow; and based on the identified application, selecting different network links from the plurality of network links for forwarding different sets of packets in the packet flow through the WAN; wherein the selection of the different network links comprises reordering out-of-sequence packets upon arrival at a destination device using a global data sequence number (GDSN), and error correction being performed by retransmitting missing packets based on a last-seen GDSN value exchanged between peers.

12. The non-transitory machine readable medium of claim 11, wherein an edge device at the site selects the different network links.

13. The non-transitory machine readable medium of claim 11, wherein the program further comprises a set of instructions for aggregating bandwidth from the plurality of network links with a multipath transport layer protocol.

14. The non-transitory machine readable medium of claim 13, wherein the multipath transport layer protocol performs the selection of the different network links for the different sets of packets in the packet flow.

15. The non-transitory machine readable medium of claim 14, wherein the multipath transport layer protocol uses an identity of the application to perform the selection.

16. The non-transitory machine readable medium of claim 11, wherein the set of instructions for said selecting comprises a set of instructions for load balancing the different sets of packets across different selected links.

17. The non-transitory machine readable medium of claim 16, wherein the set of instructions for load balancing is an application-aware load balancing.

18. The non-transitory machine readable medium of claim 16, wherein the set of instructions for load balancing traffic comprises a set of instructions for assessing a quality of a network link based on an application type of the identified application by measuring a quality of each network link based on at least one of latency, jitter, loss, and available bandwidth.

19. The non-transitory machine readable medium of claim 11, wherein the different selected network links comprises all network links in the plurality of links.

20. The non-transitory machine readable medium of claim 11, wherein the different selected network links do not include all the network links in the plurality of links.
